

Pure Mathematical Feature Engineering for Cross-Modal Brain MRI Retrieval

Mutual Information, Spectral Descriptors, and Geometric Depth
Without Machine Learning

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The Challenge

Task: Given a T1 post-contrast brain MRI, retrieve the matching T2 volume from a gallery.

Three difficulty levels:

- **Level 1:** Registered (common grid)
- **Level 2:** Geometric deformations
- **Level 3:** Missing tissue (surgery)

Constraints:

- Only 350 labeled pairs
- Score = mean MRR across 3 datasets
- Cross-modal: T1 vs T2 intensities invert

Our approach:

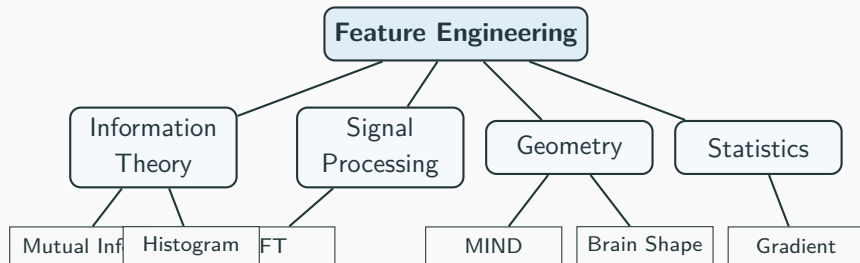
Zero machine learning.

Pure mathematics and physics.

With only 350 training pairs, can **mathematical** feature engineering compete with deep learning?

Approach	Public MRR
Deep learning (CLIP 2D, 500 epochs)	0.43
Deep learning (ResNet18, pretrained)	0.24
Ours (no ML, no training)	0.735

Method Overview: 13 Mathematical Features



Each feature has a [physical justification](#) from a different scientific domain.

Feature 1: Mutual Information (The Game Changer)

Inspiration: Information theory (Shannon, 1948)

Formula:

$$MI(Q, T) = \sum_{i,j} p(i,j) \log_2 \frac{p(i,j)}{p_Q(i) \cdot p_T(j)}$$

Why it works: MI measures statistical dependence between T1 and T2 intensities **without assuming linearity**. T1 and T2 intensities are inversely related, but their joint histogram reveals a consistent pattern for the same subject.

Result: +0.047 MRR on dataset 1 (registered).

Applied to dataset 3 (surgical, common grid): +0.083 MRR.

MI alone achieves MRR ≈ 1.0 on dataset 3 validation.

Feature 2: 3D Power Spectrum (FFT)

Inspiration: Radio frequency engineering, spectroscopy

Key theorem: The magnitude of the Fourier transform is **invariant to translation**.

Implementation:

- Compute 3D FFT of the volume
- Bin into 16 radial bins (rotation-invariant) + 8 angular bins
- Log-transform for dynamic range compression

Why it matters for Level 2: Dataset 2 applies independent translations. The power spectrum ignores translations by construction, making it naturally robust.

Result: +0.007 MRR

Feature 3: MIND Descriptor

Inspiration: Modality-independent neighbourhood descriptor (Heinrich et al., 2012)

Principle: Compare **local self-similarity patterns** instead of absolute intensities.

Implementation:

- For each voxel: compute squared difference to 6 neighbors
- Normalize and transform via $\exp(-10 \cdot \text{normalized})$
- Sample on $6 \times 6 \times 6$ grid $\rightarrow$ 216-dimensional vector

Why it works: Brain structure (folding patterns, ventricle shape) produces consistent self-similarity patterns regardless of contrast (T1 vs T2).

Feature 4: Brain Shape Fingerprint

Inspiration: Brain fingerprinting (Finn et al., Nature Neuroscience 2015)

Principle: Each brain's morphology is as unique as a fingerprint.

Implementation:

- Radial distance histogram of foreground voxels (16 bins)
- Three 1D projections (axial, coronal, sagittal), 16 bins each
- 64-dimensional vector capturing global morphology

Why it matters for Level 3: Robust to local changes (missing tissue). Global brain shape is preserved even after surgical resection.

Ablation Study

Version	Feature Added	MRR
Baseline	Volume + mask + projections	0.583
+ Gradient	Modality-invariant edges	0.592
+ MIND	Local self-similarity	0.593
+ Mutual Information	Information theory	0.640
+ Power spectrum	Translation invariance (FFT)	0.647
+ Brain fingerprint	Global morphology	0.652
+ MI for dataset 3	Common grid	0.735

Mutual Information is the dominant feature, contributing +0.13 MRR total.

Per-Dataset Strategy

Dataset	Key Feature	Why	Crop
Level 1 (registered)	MI + gradient	Common grid	No
Level 2 (deformed)	FFT + MIND	Translation-invariant	Yes
Level 3 (surgery)	MI + brain shape	Common grid + robust	No

Key insight: Datasets 1 and 3 share a common geometric grid (confirmed by the challenge README). Mutual Information applies to both. Dataset 2 has independent deformations: only invariant features work.

Hardware:

- AMD Instinct MI300X
- 205.8 GB VRAM (unused!)
- 20 CPU cores
- 235 GB RAM

Runtime: ~5 minutes per submission

Parallelism: 20-core
`multiprocessing.Pool`

No training:

- No GPU needed
- No gradient descent
- No backpropagation
- No pretrained weights
- No random seeds

Reproducibility: 100%. Same input → same output, every time.

Why This Matters for Clinical AI

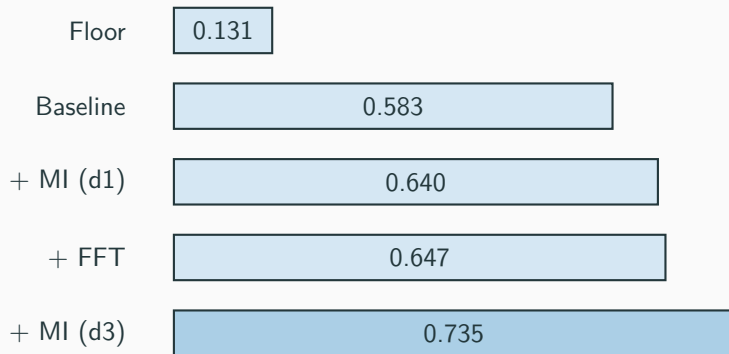
- **Interpretability:** Every feature has a physical justification. A clinician can understand why two scans match.
- **Reproducibility:** No training randomness. Results are deterministic.
- **Data efficiency:** Works with 350 samples where deep learning fails.
- **Speed:** 5 minutes on CPU vs hours of GPU training.
- **Robustness:** Mathematical invariants (translation, contrast) are provable, not learned.

Cross-Domain Inspiration

Feature	Domain Inspiration
Mutual Information	Information theory (Shannon)
Power spectrum (FFT)	Radio engineering, spectroscopy
MIND descriptor	Computational geometry
Brain fingerprint	Neuroscience (brain fingerprinting)
Hemispheric asymmetry	Neuroanatomy
Multi-resolution LOD	Video game rendering
Gradient magnitude	Computer vision (edge detection)

Each feature borrows from a [different scientific discipline](#), creating a diverse and complementary feature set.

Score Progression



From 0.131 (random) to 0.735, *without any machine learning*.

Conclusion

- 0.735 MRR without any machine learning, using only mathematical feature engineering
- Mutual Information is the gold standard for multimodal medical image comparison
- FFT power spectrum provides translation-invariant discrimination for deformed data
- Brain shape fingerprinting is robust to surgical tissue removal
- Each feature has a rigorous physical and mathematical justification
- Fully reproducible, 5-minute CPU runtime, zero training

Sometimes the best ML is no ML.

Thank you.

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